PAPER_551: U_g 26th-Order Factorial Anti-Collapse and Ug4 Dual 13+13 Split

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

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§1 Abstract

The gravitational field term U_g in UQFF is standardly written in first-order form for validation against observational datasets. The full 26th-order form, derived from 26D dimensional reduction, yields $U_{g1}^{(26)} = 26!\,a_0$ as a constant term (stable core) from the highest-degree polynomial, and $U_{g4}^{\text{split}} = (13!)^2 \cdot r \cdot t$ from the dual 13+13 splitting of $\partial^{26}(r\cdot t)$. The latter provides the BH–star duality in the UQFF: two BH26 13-mode sub-series correspond to the split. The factorial bound $26!$ establishes a vacuum-energy-level anti-collapse density threshold of $\rho_{\min} \approx 2.48 \times 10^{-30}\ \text{kg/m}^3$, below which no physical system can exist and singularities are mathematically impossible.

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§2 The 26th-Order U_g Full Form

$$U_g = g \cdot \frac{SCm}{UA}\left(U_{g1} + U_{g2} + U_{g3} + U_{g4}\right)$$

At 26th order:

$$U_{g1}^{(26)} = \frac{\partial^{26}(DPM_n \cdot SCm)}{\partial r^{26}}, \qquad U_{g4}^{\text{split}} = \frac{\partial^{13}(r \cdot t)}{\partial r^{13}} \cdot \frac{\partial^{13}(r \cdot t)}{\partial t^{13}}$$

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§3 Ug1 Stable Core from Degree-26 Polynomial

If $DPM_n \cdot SCm \approx \sum_{k=0}^{26} a_k\, r^{26-k}$ (complete degree-26 polynomial for the full 26D manifold), then applying $\partial^{26}/\partial r^{26}$:

- All terms with degree < 26 vanish (their 26th derivative is zero)
- The constant term $a_0\,r^{26}$ contributes: $\partial^{26}(a_0 r^{26})/\partial r^{26} = 26!\,a_0$

$$U_{g1}^{(26)} = 26!\,a_0 = 4.033 \times 10^{26} \cdot a_0$$

Physical interpretation: The 26th derivative isolates the stable core constant — the single surviving term confirms that the DPM gravity field has an irreducible constant baseline at 26th order, preventing any zero solution at $r > 0$.

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§4 Ug4 Dual 13+13 Split

The Ug4 term (BH tidal, PAPER_547) extends to the 26th-order split:

$$U_{g4}^{\text{split}} = \frac{\partial^{13}(r \cdot t)}{\partial r^{13}} \cdot \frac{\partial^{13}(r \cdot t)}{\partial t^{13}}$$

Computing each factor (treating $r \cdot t$ as degree-1 in each variable):

$$\frac{\partial^{13}(r \cdot t)}{\partial r^{13}} = 13!\cdot t = 6.227 \times 10^9 \cdot t$$
$$\frac{\partial^{13}(r \cdot t)}{\partial t^{13}} = 13!\cdot r = 6.227 \times 10^9 \cdot r$$

$$U_{g4}^{\text{split}} = (13!)^2 \cdot r \cdot t = 3.878 \times 10^{19} \cdot r \cdot t$$

At BH inner-scale parameters ($r = 10^{-5}\ \text{AU} = 1.496 \times 10^6\ \text{m}$, $t = -10$):

$$U_{g4}^{\text{split}} = 3.878 \times 10^{19} \times 1.496 \times 10^6 \times (-10) \approx -5.80 \times 10^{26}$$

The split is **physically motivated**: r relates to spatial DPM structure, t to temporal time-reversal dynamics. The two separate 13th derivatives represent the BH–star duality — half the 26 dimensions are spatial (r), half temporal (t).

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§5 Anti-Collapse Factorial Threshold

At the equilibrium condition $\partial^{26} F_U / \partial r^{26} = 0$:

$$26!\,g\,\frac{SCm}{UA} = \frac{\partial^{26} U_b}{\partial r^{26}}$$

Expanding $U_b = \rho\,g\,(1 - 1/\rho)$ to degree 26 in $1/\rho$, the balance condition yields:

$$\rho_{\min} > \frac{g \cdot SCm}{26! \cdot UA}$$

Numerically ($g = 10^{-3}$, $SCm = UA = 1$):

$$\rho_{\min} = \frac{10^{-3}}{4.033 \times 10^{26}} \approx 2.48 \times 10^{-30}\ \text{kg/m}^3$$

This is the vacuum energy density threshold — far below the observed density of any physical system (even the intergalactic medium at $\sim 10^{-27}\ \text{kg/m}^3$). Therefore:

$$\rho_{\text{system}} > \rho_{\min} \quad \Rightarrow \quad F_U \neq 0 \text{ at any } r > 0 \quad \Rightarrow \quad \text{No singularity}$$

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§6 Three UQFF Number Systems

System	Context in Ug26D
VDS	P_{order} couples to 26D harmonic denominator for polynomial normalisation
DVP	13+13 split → two 13-prime orbit pairs; orbits characterised by prime residues mod $p=113$
BH26	U_{g4} dual 13-mode split maps exactly to two halves of BH26 harmonic ladder (modes 1–13 and 14–26)

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§7 Conclusions

The 26th-order U_g framework:

1. **Isolates the stable constant core** $26!\,a_0$ — confirming irreducible gravity at 26th level
2. **Splits U_{g4} into BH–star duality** $(13!)^2\,r\,t$ — physically captures the temporal-spatial asymmetry of BH accretion
3. **Establishes a vacuum-level anti-collapse threshold** $\rho_{\min} = 2.48\times10^{-30}\ \text{kg/m}^3$ — every physical system density exceeds this; singularities are prohibited by the factorial factorial bound $26!$

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